

COLLISION REPAIR INFORMATION BULLETIN



SUBJECT:	PANEL BONDING ADHESIVES		
BULLETIN NO.:	TS–TCI–CRIB–2038	DATE ISSUED:	August, 2013
MODELS:	All Toyota & Scion Models	SECTION:	Structural

INTRODUCTION:

As collision repair technology continues to evolve, collision repair centers are faced with a variety of decisions regarding acceptable repair procedures. Toyota/Scion continues to set the standard by providing insight and direction when dealing with these complex issues.

The replacement of body panels using the processes known in the collision repair industry as “panel bonding” and “weld–bonding”, in conjunction with the adhesive material known as “body panel adhesive” or “structural adhesive”, is **not recognized as an approved procedure by Toyota/Scion**. Panel bonding adhesive can only be used where specified in the Repair Manuals for Collision Damage. There are specific areas that utilize panel bonding.

Refer to the appropriate Toyota/Scion Collision Repair Manuals for detailed instructions and locations of panel bonding adhesive. For Toyota or Scion Collision Centre facilities, visit the ETAS and Toyota Technical Information System (TIS) site located in the Infostream website portal at <http://infostream.toyota.ca/> to access Repair Manuals for Collision Damage and other Collision Repair Information Bulletins. For independently operated collision centres interested in these publications, visit www.techinfo.toyota.com for a subscription and access.

BODY PANEL REPLACEMENT GUIDELINES:

Toyota/Scion promotes the use of model–specific Repair Manuals for Collision Damage, which clearly indicate the procedures and specifications necessary to accomplish a Toyota/Scion approved body panel replacement. These procedures provide information on the following body panel replacement topics:

- Body panel removal & replacement specifications
- Specific cut and join locations and measurements
- Specific weld–types used to replace body panels
- Specific use of panel bonding adhesive in specified areas/locations

NOTE: Issue of this Collision Repair Information Bulletin by Toyota Canada Inc. is to ensure that proper repair information is available. It does not constitute an authorization for repair. PLEASE ROUTE THIS BULLETIN TO YOUR COLLISION REPAIR CENTER MANAGER AND COLLISION & REFINISH TECHNICIANS.

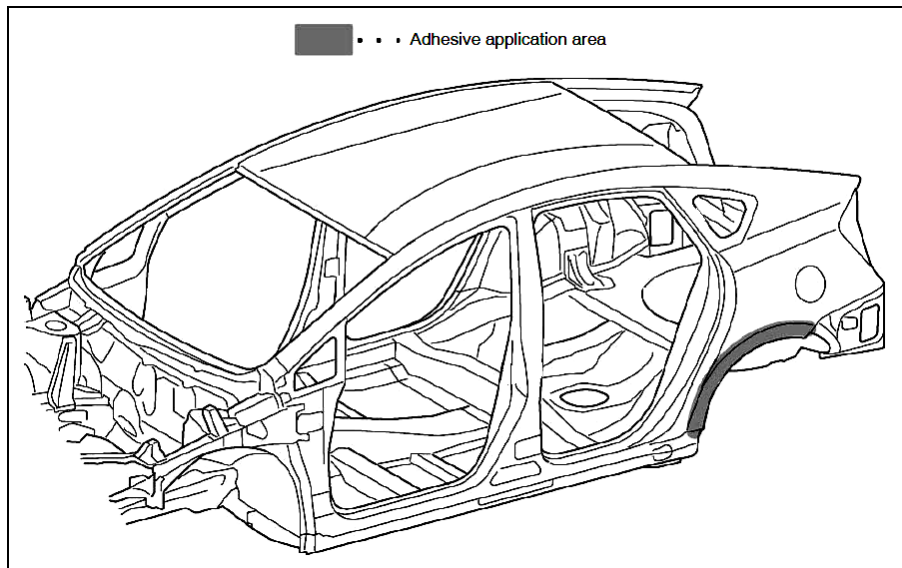


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Always refer to the Repair Manual for Collision Damage for the specific model for locations of any panel bonding adhesive use in body panels. The typical use of panel bonding adhesive on body panels is the rear quarter panel at the wheel arch flange area as shown in the example below for the 2012 Toyota Prius. For this model, the Repair Manual for Collision Damage under the “Structural Outline” section specifies the location as follows:



2012 Toyota Prius Adhesive Use on Rear Quarter Panel

NOTE: the panel bonding adhesive is typically applied 5 mm above the actual edge of the wheel well flange to provide strength, however, spot welds or MIG plus welds are still required along the flange for proper strength. **PANEL BONDING ADHESIVE DOES NOT REPLACE WELDING** during the repair – it is a supplemental bond for the panel. The practice of using only panel bonding in place of welds is not recommended as it will not provide the structural strength of the panel and overall vehicle.

During a typical rear quarter panel replacement where panel bonding adhesive is used, the panel replacement instructions will provide detailed repair procedures for removal and installation of the body panel adhesive.

Refer to the next two pages for an example of the 2012 Toyota Prius Repair Manual for Collision Damage which specifies the instructions for removal and installation of the rear quarter panel using panel bonding adhesive. **The repair for each model will vary, so always refer to the model specific Repair Manual for Collision Damage.**



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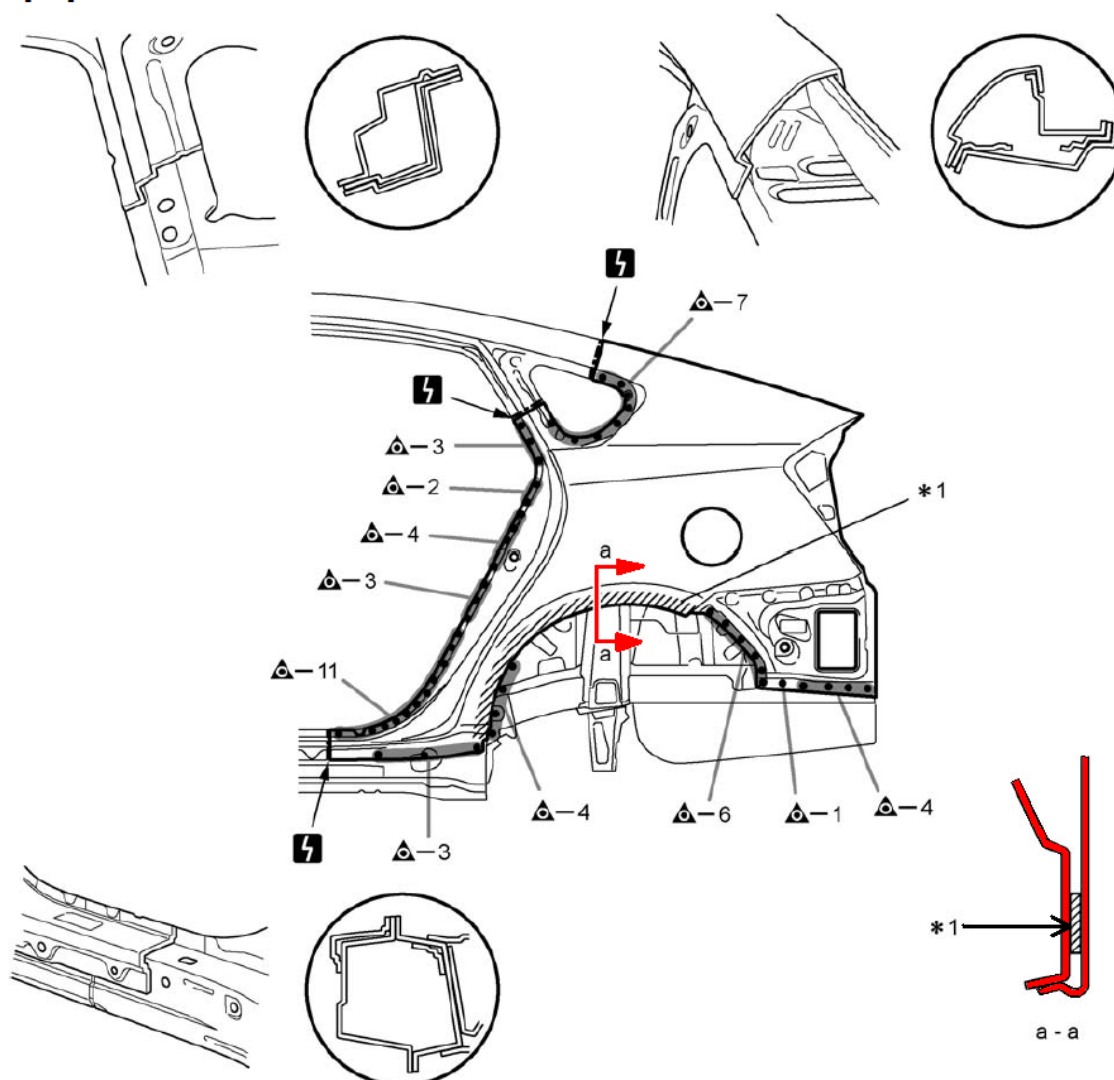
Example of Rear Quarter Panel with Panel Bonding Adhesive Removal **Instructions for 2012 Toyota Prius**

Symbol meaning

: Remove Weld Points
 : Cut with Disc Sander etc.
 : Cut Location for Supply Parts

REMOVAL

[LH]



REMOVAL POINT

- 1 *1 in illustration above indicates where the adhesive is located.
 - 2 Roughly cut open the panel so that the adhesive can be reached. Cut through the adhesive with a cut chisel to remove the panel.
- HINT:**
 In cases where the adhesive cannot be removed with a cut chisel, heat the adhesive with an industrial heater gun or gas burner taking care not to cause panel deformation by overheating.



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Example of Rear Quarter Panel with Panel Bonding Adhesive Installation Instructions for 2012 Toyota Prius

Symbol meaning ● : Spot Weld □ M I : Plug Weld : Butt Weld : Body Sealer	
INSTALLATION	
[LH] c - c *5 Spot Tip 3 (0.12) 5 (0.20) Rivets Fuel Filler Opening Lid d - d 5 (0.20) mm (in.)	
INSTALLATION POINT 1 Inspect the fitting of the related parts around the new parts before welding. This affects the appearance of the finish. 2 Temporarily install the new parts and measure each part of the new parts in accordance with the body dimension diagram. (See the body dimension diagram) 3 Before installing a new part, apply body sealer. <i>HINT:</i> • Apply body sealer evenly about 5 mm (0.20 in.) from the flange, avoiding any oozing. • Apply body sealer evenly, about 3 – 4 mm (0.12 – 0.16 in.) in diameter. 4 Apply adhesive (3M™ Automix™ Panel Bonding Adhesive #8115) to the area indicated by *3 in the illustration. <i>HINT:</i> Apply enough adhesive for the panels to stick to each other. 5 Perform spot-welding on the flange indicated by *4 in the illustration. Modify/cut the spot tip as shown in the illustration*5 so that it can fit in to the narrow flange.	



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PANEL BONDING ADHESIVES

In addition to the 3M™ Automix™ Panel Bonding Adhesive #8115 recommended in the Repair Manuals for Collision Damage, Toyota Canada Inc. makes available the equivalent Würth Power Bond (Art No. 893.450100) through the Würth Body Shop Products Program.

Toyota and Scion Dealers can order all of Würth's products through this program. Dealers can access the program details in the Infostream website portal at <http://infostream.toyota.ca/>. For independently operated collision centres interested in these products, please contact your local Toyota and Scion Dealer.



Würth Power Bond
Art. No. 893.450100